



UNIVERSITY OF PERADENIYA - FACULTY OF SCIENCE
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
END SEMESTER EXAMINATION (SEMESTER I) – 2022/2023
CSC1013/CS101 – INTRODUCTION TO COMPUTER SCIENCE
AND PROGRAMMING)



Answer ALL Questions

Time allowed: Two Hours

Instructions

- This is a closed book examination.
- Calculators are not allowed.
- There are Four (04) questions on four (04) pages.
- All questions carry equal marks.
- Clearly write the Question number and Part number in your answers.
- Make your answers concise and precise.

- (a) Explain using examples why we require different number systems. [02 marks]
 - (b) What are the logical operations performed in a computer system? [03 marks]
 - (c) Select the most appropriate answer. [20 marks]
 - i. Which of the following cannot be used for representing data in a computer system?
(a) Binary 1 and 0 (b) +Vcc and 0V (c) Decimal (d) Magnetism
 - ii. How many bits are 1 Megabyte of data?
(a) 8×2^9 (b) 2^6 (c) 2^9 (d) 2^{12}
 - iii. What is the decimal value of the 10110110_2 ?
(a) 182 (b) 100 (c) 155 (d) 134
 - iv. What is the hexadecimal representation of 10110110 ?
(a) A6 (b) C6 (c) 116 (d) B6
 - v. Which of the following is the equivalent binary value for decimal value 0.675?
(a) 0.10101 (b) 1.0101 (c) 0.10010 (d) 0.00101
 - vi. Which of the following logic gates provides an output of 1 when and only when one input is 0 and the other is 1 in a two input gate?
(a) NAND (b) XNOR (c) NOR (d) XOR
 - vii. What is the BCD value of 100100110101_2 ?
(a) 4465 (b) 935 (c) 210311 (d) 953
 - viii. Which of the following is the sign-magnitude representation of -65?
(a) 01000010 (b) 10111110 (c) 01000001 (d) 11000001
 - ix. What is the range of values that can be represented in Signed 1's Complement which has n bits?
(a) $-(2^{n-1} - 1)$ to $(2^{n-1} - 1)$ (b) $-(2^{n-1} - 1)$ to $(2^{n-1} - 1)$
(c) $-(2^{n-1})$ to $(2^{n-1} - 1)$ (d) $-(2^{n-1})$ to (2^{n-1})

- x. Which of the following is the smallest decimal number that can be represented in Signed Fixed-point $\langle 8, 2 \rangle$ number representation?
 (a) -63.75 (b) 0 (c) -32.75 (d) -31.75
2. (a) What is the decimal value of the floating point representation 1 10000110 1011010001000000000000 in IEEE 754 standard? [04 marks]
- (b) Perform the following binary arithmetic operations. Clearly indicate the steps.
- 00101101 – 01001100 Using 2's complement [02 marks]
 - 00101101 + 00001111 [02 marks]
 - 00010101 $\times$ 00000100 [02 marks]
- (c) Simplify the sum of products expression $F(A, B, C) = \Sigma(m_0, m_2, m_5, m_7)$ using Boolean algebraic laws. [05 marks]
- (d) Using Karnaugh maps simplify the following Boolean equation:
 $f(A, B, C) = (A + B + C) \cdot (\bar{A} + \bar{B} + \bar{C}) \cdot (A + \bar{B} + \bar{C}) \cdot (\bar{A} + \bar{B} + C) \cdot (A + \bar{B} + C)$
 [05 marks]
- (e) Draw the simplified equation in part d using logic gates. [02 marks]
- (f) Draw the simplified circuit using NAND and NOR gates. (*Hint: use DeMorgan's theorem*) [03 marks]
3. (a) What is a computer programme? [02 marks]
- (b) Explain briefly why we require programming languages. [04 marks]
- (c) What is the difference between a compiler and an interpreter in terms of program execution? [04 marks]
- (d) What is the difference between a syntax error and a runtime error? [04 marks]
- (e) Draw a flowchart (using standard symbols) to find the largest number among 10 user input random numbers. You may use a proper way to store numbers before processing. [08 marks]
- (f) Write pseudocode for the flowchart drawn in part e. [03 marks]
4. (a) What is the difference between Structured oriented programming & Object-oriented programming? [02 marks]
- (b) Briefly explain using examples the main concepts in object-oriented programming. [03 marks]
- (c) Write the outputs of the following C snippets. [10 marks]

[i)]

i.

```
#include <stdio.h>
int main() {
    int a = 100;
    int b = 30;
    for (int i = 1; i <= 3; i++) {
        a += i;
        b -= i;
    }
    printf("%d\n", a);
    printf("%d\n", b);
    return 0;
}
```

ii)

ii.

```
#include <stdio.h>
int main() {
    int x = 0;
    while (x < 20) {
        if (x % 2 == 0) {
            x++;
        } else {
            x += 5;
        }
    }
    printf("%d\n", x);
    return 0;
}
```

- (d) Write a C program `reverse.c` to reverse the order of integer numbers available in an array with ten elements using same array. [10 marks]